

Deep Reinforcement Learning for Congestion Control and Routing Optimization in LEO Satellite Networks

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Abstract—This paper provides a exploration of the application of low earth orbit satellite networks (LSN) in the context of sixth-generation(6G) mobile communication technology. The primary focus of this research is to address issues related to congestion control and routing within satellite networks. We introduces ECSAC-DRO, a novel algorithm that leverages deep reinforcement learning (DRL) to optimize routing and congestion control. Experimental results demonstrate that the ECSAC-DRO algorithm, taking into consideration the high-dimensional characteristics of LSN and incorporating elements from the DRL-based Soft Actor-Critic (SAC) algorithm, excels in providing flexible control over LSN congestion levels and bandwidth utilization. It outperforms traditional methods like Dijkstra, making it a promising and forward-looking solution that lends support to the future development of 6G communication technology.

Keywords—Congestion control, LEO satellite networks routing, deep reinforcement learning

I. INTRODUCTION

The advent of 5G has advanced the Internet of Things, but challenges persist, prompting anticipation for 6G [1]. 5G's network communication is limited by hardware, and its global deployment demands substantial resources. 6G strives for global coverage with a satellite network focus, making communication technology and integration its central research concerns [2].

The existing satellite orbits primarily comprise Low Earth Orbit (LEO), Medium Earth Orbit (MEO), and Geosynchronous Transfer Orbit (GTO). As technology continues to advance, the success rate of satellite launches is continuously improving, costs are decreasing, and LEO orbit resources are more abundant [3]. However, to fully realize their potential, LEO orbits require a large number of satellites to form satellite constellations. In recent years, the development of large-scale constellation networks has been rapid. As these networks continue to evolve, the issues they face are gradually turning into pain points. Specifically, due to geographical, political, and other factors, congestion near the traffic entry and exit points of constellation routing networks has become a serious concern. This congestion is increasingly becoming a bottleneck for the overall network and may have disastrous consequences for the entire network infrastructure[4]-[6].

Traditional approaches, such as employing distributed routing concepts, involve recalculating routes when

performance bottlenecks occur at intermediate nodes[7]. However, given the high dynamism and complexity of LEO satellite constellation networks, route recalculations result in a significant computational burden with limited efficiency gains. While optimizing algorithms can be used to reduce the computational load, this often overlooks the predictability and structured nature of satellite orbits, despite their dynamic characteristics.

Another route for resolution involves adopting a broadcasting method to seek optimal paths [8], but this method leads to substantial resource and energy wastage. Furthermore, the extensive routing calculations associated with LEO constellations make convergence challenging. In response to these challenges, researchers have explored strategies such as employing the nearest neighbor propagation clustering algorithm to solve the problem of uneven distribution of ground users, fluctuating reliability of nodes and links, and frequent switching of controller groups [9]. They have also explored heuristic algorithms tailored to specific regions, aiming to mitigate congestion issues [10]. However, these methods heavily rely on manual configuration and exhibit limited robustness, hindering their widespread adoption.

In recent years, the field of machine learning has experienced a rapid evolution, with DRL emerging as a convergence of deep learning's robust cognitive capabilities and reinforcement learning's decision-making prowess. DRL empowers intelligent agents to explore their surroundings within predefined contexts, quantifying the consequences of their actions and assessing their navigational outcomes. This process culminates in the formulation of strategies for optimal decision-making in complex environments. Consequently, DRL has found broad-ranging applications across domains like natural language processing, computer vision, and autonomous driving[11]-[13]. Likewise, DRL holds the potential to make significant contributions in the realm of LEO route optimization.[14] turned to a DQN-based algorithm to tackle the load balancing issues within LEO routing networks. However, a drawback of this approach is the discrete action space, which limits its capacity to faithfully represent real-world challenges. What's more, [15] ventured into the realm of using a DDPG-based algorithm to combat the issue of excessive energy consumption in LSN. Employing a continuous action space, this innovative approach offers a fresh perspective on alleviating routing congestion within LSN. Nevertheless, DDPG's performance in high-dimensional state spaces left much room for

improvement.

In this context, the relatively underutilized SAC algorithm, designed for high-dimensional state spaces, emerges as a promising alternative. Hence, to address the joint optimization challenge of traffic control and routing in vast, dynamic, and predictable satellite networks, our plan is to harness the SAC algorithm within the framework of DRL. This strategy aims to enhance the cognitive capabilities of satellite networks concerning routing and traffic distribution, ultimately paving the way for the identification of optimal decision-making.

II. SYSTEM MODEL

A. LSN Model

The satellite constellation is designed for seamless global coverage through the use of a polar orbit configuration. In this arrangement, satellites are evenly distributed within a 180-degree range along the equator. Within a single time slot, satellites in the polar orbit constellation have the ability to establish both intra-orbit and inter-orbit connections, which are represented as a graph $G = (V, L)$. In this context, $V = \{v_1, v_2, \dots, v_n\}$ signifies the set of satellites, while L represents the collection of interconnections between these satellites.

When numbering the orbits from 1 to n , with i ($1 < i < n$) denoting a specific orbit, satellites on the i -th orbit can establish connections with up to two adjacent satellites on the same orbit, and up to two adjacent orbits, namely, the $(i-1)$ -th and $(i+1)$ -th orbits. However, it's essential to note that due to the presence of a seam or boundary at the equator, satellites on the 1st orbit are unable to establish connections with satellites on the n -th orbit. As a result, the spatial distribution of satellites within a single time slot can be conveniently expressed by organizing V in matrix form.

$$V = \begin{bmatrix} v_{1,1} & v_{1,2} & v_{1,3} & \dots & v_{1,j} \\ v_{2,1} & v_{2,2} & v_{2,3} & \dots & v_{2,j} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ v_{i,1} & v_{i,2} & v_{i,3} & \dots & v_{i,j} \end{bmatrix} \quad (1)$$

where $v_{i,j}$ represents the presence or existence of satellites in the j -th region of the i -th orbit. $v_{i,j} = 0$ means there is no satellite in j -th region of the i -th orbit. $v_{i,j} = 1$ means there is satellite in j -th region of the i -th orbit.

The interconnections among satellites, denoted by the matrix L , can also be depicted as an n -by- n matrix.

$$L = \begin{bmatrix} l_{1,1} & l_{1,2} & l_{1,3} & \dots & l_{1,n} \\ l_{2,1} & l_{2,2} & l_{2,3} & \dots & l_{2,n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ l_{n,1} & l_{n,2} & l_{n,3} & \dots & l_{n,n} \end{bmatrix} \quad (2)$$

Within this matrix, each element signifies the direct connectivity between the m -th and n -th satellites. $l_{mn} = 0$ means the m -th satellite is not connected to the n -th satellite. $l_{mn} = 1$ means the m -th satellite is connected to the n -th satellite.

B. Problem Formulation

When data packet i needs to travel from the source satellite to the destination satellite, the transmission path can be denoted as path:

$$p_i^{v_s, v_d} = \{p_i^{v_s, v_n}, p_i^{v_s, v_{n-1}}, p_i^{v_s, v_{n-2}}, \dots, p_i^{v_{m-1}, v_d}, p_i^{v_m, v_d}\} \quad (3)$$

At a specific time t , when there is a total of p data packets to be transmitted from source nodes to destination nodes, each data packet has the autonomy to choose a path from the available connections in L for its transmission. Consequently, when simultaneously transmitting p data packets, the distribution of data packets within the network can be summarized as follows:

$$P = \sum_{i=0}^p p_i^{v_s, v_d} = \begin{bmatrix} p_{1,1} & p_{1,2} & p_{1,3} & \dots & p_{1,n} \\ p_{2,1} & p_{2,2} & p_{2,3} & \dots & p_{2,n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ p_{n,1} & p_{n,2} & p_{n,3} & \dots & p_{n,n} \end{bmatrix} \quad (4)$$

The expression for the average bandwidth utilization of a satellite, assuming that an ISL requires a bandwidth of k to transmit a single data packet, can be represented as follows:

$$\eta = \sum_{i=0}^n \frac{kp}{B}, \forall p \neq 0 \quad (5)$$

At time t , we define the instantaneous congestion function of the network, which is used to measure the degree of congestion in the network at that particular moment.

$$F_t(p) = \sum_{i=0}^n \sum_{j=0}^n \delta^{p_{i,j}}, \delta > 1 \quad (6)$$

Here, δ stands for the route convergence control factor. So, the routing optimization problem under study can be represented as a non-linear programming problem :

$$Q = \min_p F_t(p) \quad (7)$$

$$\text{s.t. } \eta > \zeta$$

The objective of this problem is to maximize the reduction of network congestion at the given average satellite bandwidth utilization ζ while maintaining.

III. EVENT-DRIVEN CONGESTION-AWARE SAC-ENHANCED DIJKSTRA ROUTING OPTIMIZATION (ECSAC-DRO)

When it comes to route selection in a pre-defined satellite constellation plan, the algorithm can make route choices for each event based on the current network conditions. These choices aim to fulfill congestion requirements and minimize resource wastage, including energy and bandwidth. The definitions for the state, action, and reward functions of the agent when handling events from the source node to the destination node are outlined below.

A. DRL Model

1) State Space

The initial state space comprises an n by n matrix, represented as S . As each data packet selects its transmission route from the source node to the destination node, the corresponding element in the matrix S increments by 1. Consequently, when all p data packets have designated their transmission routes at a specific moment, the matrix S effectively portrays the present state of routing path selections

within the satellite network, illustrating the distribution of traffic.

$$S = \begin{bmatrix} s_{1,1} & s_{1,2} & s_{1,3} & \cdots & s_{1,n} \\ s_{2,1} & s_{2,2} & s_{2,3} & \cdots & s_{2,n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ s_{n,1} & s_{n,2} & s_{n,3} & \cdots & s_{n,n} \end{bmatrix} \quad (8)$$

2) Action Space

The action space determines the traffic distribution of the next data packet to be transmitted, as provided by the agent.

$$A_i^{v_s, v_d} = \begin{bmatrix} a_{1,1} & a_{1,2} & a_{1,3} & \cdots & a_{1,n} \\ a_{2,1} & a_{2,2} & a_{2,3} & \cdots & a_{2,n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{n,1} & a_{n,2} & a_{n,3} & \cdots & a_{n,n} \end{bmatrix} \quad (9)$$

Here, 'i' represents the i-th event, and 'a' denotes the data packet distribution to be carried by satellites from the x-th to the y-th satellites.

3) Reward

Upon the successful transmission of all data packets and the completion of the task, the reward for this exploration is calculated using the following formula:

$$R = \sum_i \delta^{s_i} - \sum_j \delta^{s_j} + \varepsilon \quad (10)$$

$$, \forall i \in \mathcal{L}', \forall j \in \mathcal{L}, i, j \neq \{n_s, n_d\}$$

$\mathcal{L}'$ represents the collection of paths obtained through the Dijkstra shortest path algorithm, while $\mathcal{L}$ comprises the set of routing paths chosen for the current scenario. In this context, s_i signifies the state value at the 'i' location within the state matrix, ε serves as the offset control factor that determines the threshold between rewards and penalties. The values near the destination in the state matrix, reflecting higher congestion due to a significant number of data packets being transmitted, result in a smaller R value, which may even be negative, thereby leading to penalties.

B. Algorithm Flow

When the routing management system notifies the agent that routing event i is commencing, originating from the source node v_s and concluding at the destination node v_d , the agent makes decisions based on the current network state, which reflects the existing traffic distribution within the network. This process leads to the determination of the routing distribution for this specific transmission, referred to as action $A_i^{v_s, v_d}$. Subsequently, the routing management system calculates the reward for this event's transmission. This calculation takes into account the effect on the state S as provided by the agent and employs the Dijkstra algorithm. The agent then receives this reward, updates its neural network, and adjusts S to the new state S_{t+1} , which has been modified by $A_i^{v_s, v_d}$.

In our routing management system, we utilize the SAC algorithm from DRL. SAC combines deep neural networks and policy gradient methods to facilitate learning and optimization. It excels in handling not only continuous state and action spaces but also offers improved convergence properties compared to conventional Q-learning methods.

Furthermore, SAC introduces action entropy to strike a balance between exploration and exploitation, preventing premature policy convergence to local optima.

The specific process of the algorithm is shown in Algorithm 1.

Algorithm 1 ECSAC-DRO

Input: LSN matrix.

Output: The routing policy for the transmission events.

Initialize Critic network Q_{ϕ_1}, Q_{ϕ_2} , Actor network π_{θ} and pool D.random parameters ϕ_1, ϕ_2, θ , target network $\phi_1, \phi_{\text{targ},1} \leftarrow \phi_1, \phi_2, \phi_{\text{targ},2} \leftarrow \phi_2, \theta' \leftarrow \theta$

for episode < M **do**:

Obtaining the L and S_0 through the S.

for i = 1 to I **do**:

action $A_i^{v_s, v_d} \sim \pi_{\theta}(s) + \varepsilon, \varepsilon \sim N(0, \sigma)$

$s', \text{reward}, \text{done} = \text{step}(\text{action})$

$s', \text{reward}, (s, a, r, s', \text{done})$ store in D

randomly draw N samples from D in batches (s, a, r, s', done)

update Critic network Q_{ϕ_1}, Q_{ϕ_2}

update Actor network π_{θ}

update target network ϕ_1, ϕ_2

end for

end for

IV. PERFORMANCE EVALUATION

A. Experimental Environment Configuration

We have developed a Python-based computational environment for satellite constellation network routing. Additionally, we have constructed a DRL algorithm model using Python and conducted an extensive series of simulation evaluations for the ECSAC-DRO algorithm. The experiments were performed on a network with a satellite position matrix randomly generated, consisting of 20 satellites. We established random connection relationships and transmitted 50 data packets. Prior to exploration, each generated network underwent an assessment to determine its validity, ensuring that bad cases had minimal impact on the experiments. The primary goal of the DRL exploration in this study is to select routing paths for data packet transmission based on congestion levels and average bandwidth utilization. The aim is to optimize network congestion and average bandwidth utilization, striving to reach an ideal level.

B. Analysis of Experimental Results

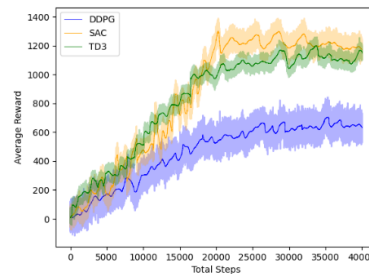


Fig. 1. Learning curve of average reward in the proposed ECSAC-DRO with different arithmetics.

As shown in Fig. 1, we compared the ECSAC-DRO algorithm with DRL algorithms using DDPG and TD3 in terms of convergence speed and average reward when

performing tasks. Our analysis reveals that DDPG exhibits the slowest convergence speed, with its learning curve being more susceptible to fluctuations. The achievable average reward value is also relatively lower. This is because, while DDPG performs well in continuous state spaces, it requires more samples for learning in larger state spaces and can become trapped in local optima. In contrast, TD3 demonstrates greater stability with relatively minor fluctuations during the early exploration stage and a faster convergence speed. This can be attributed to various improvements that help reduce overestimation errors during training. The ECSAC-DRO algorithm places a higher emphasis on exploration. Although it converges slower compared to TD3 and exhibits more pronounced fluctuations during the early training phase, it

ultimately yields slightly higher final rewards than TD3.

When comparing the ECSAC-DRO algorithm with random routing and the Dijkstra algorithm in terms of average latency, actual bandwidth utilization, and overall network congestion, our analysis, as shown in Fig 2, indicates that the Dijkstra algorithm achieves the lowest average latency but incurs the highest congestion when used. The random routing algorithm results in the longest average latency and lower actual bandwidth utilization. On the other hand, the ECSAC-DRO algorithm, while exhibiting slightly longer latency compared to Dijkstra, significantly outperforms random routing in terms of latency and actual bandwidth utilization.

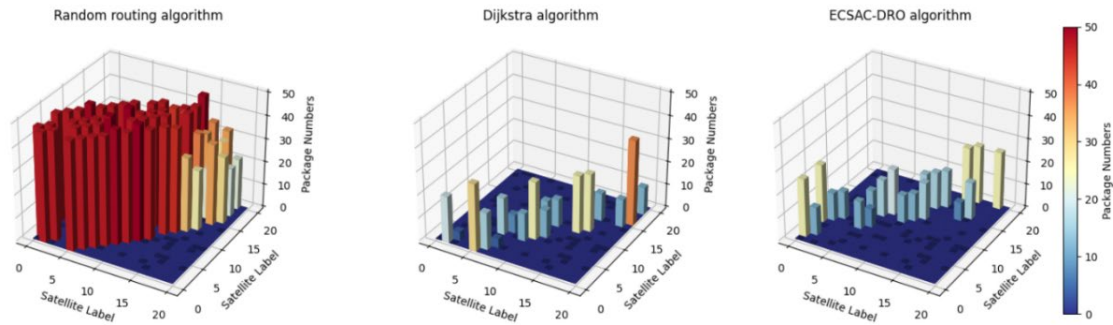


Fig. 2. LSN states with different algorithms

V. CONCLUSION

In summary, the experimental results underscore the significant boost in network performance that ECSAC-DRO algorithm brings to LSN. It offers fine-tuned control over congestion levels and bandwidth utilization. When compared to traditional algorithms, particularly random routing, these algorithms prove to be highly competitive. These findings provide robust support for LSN performance optimization and routing decision-making.

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